

ExamForge AI

Deployment Guide

Phase 6 — Production Deployment

August 2, 2026

Deployment Guide

This guide provides step-by-step instructions for deploying ExamForge AI to production.

1. Prerequisites

- Node.js 20+ and npm/bun
- Supabase project with RLS policies configured
- Flutterwave account for billing integration
- Domain with SSL certificate
- CDN/reverse proxy (Caddy, Nginx, or Cloudflare) for rate limiting

2. Environment Variables

- Copy .env.example to .env.local and fill in all required values
- NEXT_PUBLIC_SUPABASE_URL — Your Supabase project URL
- NEXT_PUBLIC_SUPABASE_ANON_KEY — Your Supabase anon key
- SUPABASE_SERVICE_ROLE_KEY — Required for webhook processing
- FLUTTERWAVE_PUBLIC_KEY — Your Flutterwave public key
- FLUTTERWAVE_SECRET_KEY — Your Flutterwave secret key
- FLUTTERWAVE_WEBHOOK_SECRET — Your webhook secret hash
- NEXT_PUBLIC_APP_URL — Your production domain URL

3. Supabase Configuration

- Enable Row Level Security (RLS) on all tables
- Configure RLS policies: profiles scoped by school_id, exams scoped by school_id/created_by
- Configure Storage policies: avatars readable by authenticated users, writable by owner only
- Enable Realtime for notifications table
- Deploy Edge Functions: ai-complete, ai-stream, flutterwave-checkout, process-refund, marketplace-download

4. Build and Deploy

- Run: npm run build
- Verify: 0 build errors, 0 TypeScript errors
- Deploy: Use standalone output with Docker or PM2
- Configure reverse proxy with HTTPS (required for HSTS)
- Set up rate limiting at CDN/proxy level

5. Post-Deployment Verification

- Verify all 5 roles can log in and access their dashboards
- Verify data isolation: school_admin cannot see other schools' data
- Verify security headers are present (check with browser dev tools)
- Verify CSP is not blocking legitimate resources
- Test all API routes return proper error responses
- Test webhook endpoint with Flutterwave test events

6. Monitoring

- Connect Sentry for error tracking (set SENTRY_DSN)
- Connect OpenTelemetry for observability
- Set up uptime monitoring
- Configure log aggregation (enterprise logger outputs structured JSON)